

Improving SMART learning: Course completion via AI-driven hybrid system integration in big data

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ARTICLE INFO

Keywords:

Smart education
Hybrid system
Artificial intelligence (AI)
Machine learning (ML)
ICT
Big data
e-learning
Learning experiences

ABSTRACT

Rapid data growth in various sectors, like education, is driving the need for robust systems capable of enhancing the learning experience and at the same time guaranteeing course completion. Integrating artificial intelligence in hybrid systems for big data management and processing is highly pertinent within smart-learning. This dissertation is about the development of an AI-enabled hybrid system that smartly leverages big data with the objective of optimizing course completion. The hybrid system implements state-of-the-art machine learning modeling techniques, including Decision Trees, Support Vector Machines, and Naïve Bayesian Classifiers for analyzing the performance of the students' data as well as for predicting successful completion of the course. In essence, when these systems are integrated, the hybrid system can accommodate the complexity and variability of educational data. Experimental analysis illustrates that the proposed model is competitive in the domain of course completion and provides a high number of accuracies, thereby enhancing its ability to boost the learning experience and academic outcomes in smart educational environments.

1. Introduction

Various reviews of literature suggest that research on large sources of educational data is critical to the enhancement of learning experiences and the enabling of students to finish a course in the field of smart learning [1]. The rapid growth of educational data is characterized by increasing volume, velocity, and variety, posing serious challenges to traditional analytic methods [2]. It could very well be that MBD methodology is not sufficiently positioned to adequately tackle the problem and amount of big data; hence they offer more advanced means of using machine learning and artificial intelligence to give some sense of insight into the big data problem [3]. Effective use of data is urgently crucial to improving learning processes and supporting students throughout their academic journeys. Supervised learning algorithms can thus facilitate the training of models on historical data to highlight variables associated with academic success and course completion [4]. In contrast, unsupervised learning approaches such as clustering approaches can identify trends and patterns without requiring labeled data. Deep learning techniques are gaining interest in this area as they have shown great

promise in the analysis of complex educational data and improving learning outcomes. An effective approach requires expertise in both education and the data analysis techniques employed.

The prediction analytics model developed in the study called Learning Record Customer (LRC)-is a supervised machine learning approach for predicting the time of completion based on frequently used parameters in the Learning Record Store (LRS). It uses predictive analytics to arrive at Completion Course Time (CCT), providing a data-backed approach to measure and improve students' learning paths. The hybrid framework methodology we propose has a structured methodology that comprises four areas: data collection, preprocessing, analysis, and modeling-individual stage is to afford relevant educational data an efficient way of dealing with prediction. Until such factors were identified by predictive analytics has a more proactive intervention for improving such a situation, boosting student engagement and retention. The model's integration in the big data ecosystem allows for real-time analysis and changes of learning paths based on acquired knowledge about individual learners to improve the learners' academic performance and learning outcome.

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<https://doi.org/10.1016/j.teler.2025.100199>

Received 5 December 2024; Received in revised form 1 March 2025; Accepted 3 April 2025

Available online 11 April 2025

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This research proposes an AI-driven hybrid model acting as a catalyst in optimizing smart learning by significantly improving completion rates of specific courses based on big data advanced data integration environment. The hybrid combines three dissimilar machine learning algorithms [5]: naive Bayesian Classifier, support vector machine (SVM), and decision tree. All three play a vital data-analytic role: the Decision Tree segregates the data according to features, the SVM refines the model whereby normal data points and variations are distinguished, while Naive Bayesian Classifier provides a probabilistic framework to estimate results. The hybridization of these three algorithms engenders the learner predictive model commonly targeted at augmenting learning experience and minimizing course completion time in smart educational environments [6].

The rest of the article is an overview. The “Related Work” section describes previous research and basic principles. The literature review discusses information theory extensively and examines Decision Tree, Support Vector Machine, and Naive Bayesian Classifier as the core methodologies in this approach. Moreover, the section on “Proposed Smart Learning Optimization Model” includes its architecture and methodology, which consists of integrated smart learning architecture, data sources identification, preparation, and pre-processing, followed by “Exploratory Data Analysis (EDA)”, culminating in proposing a framework of the hybrid system as an efficient enhancer towards learning experience optimization and course completion. The “Experimental Analysis and Evaluation” section presents the performance and effectiveness of the proposed model and concludes with future research directions in the “Conclusions” section.

2. Related works

In the context of smart learning, optimizing learning experiences and ensuring course completion requires analyzing large amounts of educational data. Base of motivation for the research is the identify real strong patterns and trends within the context of the large-scale, diverse educational data sets that exist and are expected to take a stronghold on the student learning results and therefore academic success. Numbers of approaches to tackle the nature of educational data in terms of unique challenges exerted by volume, velocity, and variety have originated. Such will hence undertake an overview of notable research and developments in data analytics and machine learning concerning educational applications. Deep learning approaches, graph approaches, scalability algorithms, and ensemble learning have become emerging focus areas. Each mode of such provides significant insights into fostering enhanced educational outcomes and optimized course completion. Many machine learning techniques for analyzing educational data have been highlighted in recent studies.

In this respect, models such as generative adversarial networks and autoencoders have been shown to contribute tremendously in recognizing hidden patterns in educational data [7]. Their investigation provides valuable perspectives for identifying underlying factors that influence learning outcomes.

Chinta [8] provided a comprehensive overview of traditional and innovative data analysis methods, addressing scalability and efficiency issues that are critical to dealing with large educational datasets [9]. studied graph-based techniques and clustering methods for analyzing complex educational networks and interactions, and showed how these methods can reveal important trends in student behavior and learning patterns [10]. addressed the scalability challenge of analyzing large-scale educational data by employing distributed computing frameworks and parallel algorithms, which are essential for managing real-time data streams and maintaining efficiency [11]. investigated the effectiveness of ensemble learning approaches in improving the accuracy of educational data analysis across different datasets and illustrated the benefits of combining multiple analysis models [12]. emphasized the need for adaptable and scalable solutions in the face of increasing data sources and diversity, and emphasized the importance of flexible

methods for analyzing educational data [13]. Studied machine learning methods tailored to specific challenges in educational data and recognized the unique characteristics and difficulties presented by different learning environments.

[14] proposed scalable methods for data analysis by focusing on efficient algorithms capable of processing large amounts of educational data with minimal computational effort [15]. Emphasized the importance of adapting existing approaches to the specific needs of educational data analysis [16]. Conducted a comprehensive review of large-scale data analysis techniques and discussed recent advances and their relevance to various educational contexts. This work identifies how to apply deep learning algorithms to reveal complex relationships and identify patterns in educational data for actionable insights on student performance and course completion [17]. Overall, advanced analytics methodologies provide a more comprehensive and critical understanding of smart learning; consequently, improved learner outcomes could be achieved through the data-efficient decision-making process.

3. Literature review

3.1. Theory of information

Information-theoretical measures provide sound basis for the analysis of educational data and the enhancement of learning deliverables. Such measures assist in business quantification processes, besides being instruments for understanding relationships within educational data sets concerning student performance and course completion rates [18]. Among those measurements, some are employed as follows:

- **Entropy:** Entropy measures the uncertainty or randomness associated with a random variable. In an educational setting, it can be used to understand the normal distribution for student performance metrics [19]. Generally, anomalies or deviations from expected levels of performance can have distinctive statistical properties, defining them with higher degrees of disorder in comparison with those that obey normal distributions. For instance, during the evaluation of student grades or engagement, unusual patterns signal areas requiring intervention or support. For a discrete random variable X with probability distribution $P(X)$, entropy $H(X)$ is defined as:

$$H(X) = -\sum_{x \in X} P(x) \log P(x) \quad (1)$$

where $H(X)$ denotes the entropy for the random variable X . $P(X)$ yields the probability of x in the space X . Logarithm can work with either base 2 (for bits) or base e (for nats).

- **Conditional Entropy:** Conditional Entropy [20] $H(X|Y)$ evaluates the uncertainty associated with variable X given information about variable Y . In terms of educational data, such a concept would indicate how much uncertainty remains about student's performance in X when additional information, for example, previous academic history Y , is known. It is defined as:

$$H(X|Y) = -\sum_{x \in X} \sum_{y \in Y} P(x, y) \log \frac{P(x, y)}{P(y)} \quad (2)$$

where $P(x, y)$ represents the joint probability distribution of X and Y , and $P(y)$ represents the marginal probability distribution of Y .

- **Relative Entropy (KL Divergence):** Relative entropy, or Kullback-Leibler (KL) divergence [21], quantifies the difference between two probability distributions. In educational settings, this measure can be

used to compare the expected distribution of student performance P with the observed distribution Q . It is defined as:

$$DKL(P \parallel Q) = \sum_{x \in X} P(x) \log \frac{P(x)}{Q(x)} \quad (3)$$

where:

$DKL(P \parallel Q)$ is the Kullback-Leibler divergence between distributions P and Q .

$P(x)$ and $Q(x)$ are the probabilities of the value x under the distributions P and Q , respectively.

The sum runs over all possible values $x \in X$, where X is the set of possible outcomes.

- **Information Gain:** Information gain [22] $IG(Y, X)$ measures the reduction in uncertainty about the target variable Y when the value of feature X is known. For educational data, this could represent how much knowing a student's study habits X reduces the uncertainty about their final grades Y . It is calculated as:

$$IG(Y, X) = H(Y) - H(Y|X) \quad (4)$$

where $H(Y)$ is the entropy of Y without knowledge of X , and $H(Y|X)$ is the conditional entropy of Y given X .

- **Information Cost:** Information cost reflects the divergence from expected behavior. In the context of SMART education, this can be interpreted as the difference between expected academic performance distributions and observed outcomes [23]. It is defined as:

$$\text{Information Cost} = DKL(P \parallel Q) \quad (5)$$

where P denotes the expected distribution - normal academic performance, whereas Q represents the observed one-module anomalous performance.

3.2. Decision tree

Decision Trees play an important role for optimization of learning experiences and completion of courses in smart learning [24]. They assist a bit during construction of complicated decision frameworks that classify educational data based on pre-defined criteria. Due to the nature of educational data, whether numerical or categorical, Decision Trees play an important role for preprocessing and initial features selection. It becomes handy to identify at-risk students or predict graduation rates through clustering of data using some significant grouping based on academic performance and engagement metrics.

3.3. Support vector machine

Support Vector Machines provide critical input to the development of the DT-SVM-NBC hybrid system for smart educational applications [25]. The main asset of the algorithm is the maximization of distances between classes, which is very helpful on high-dimensional education datasets. SVMs are mainly used for binary classification tasks, allowing them to discriminate between different levels of student performance or to predict course completion. By using kernel functions, SVMs can handle complex nonlinear relationships in education data and improve the accuracy of predictions of student success and the identification of outliers that may require additional support [26].

3.4. Naive Bayesian classifier

The Naive Bayesian Classifier stands out as the other main piece engaging in the DT-SVM-NBC hybrid system for intelligent learners [27]. Naive Bayesian classifiers have been able to use the assumption of independence between features given the class label to simplify the probabilistic model, thus making it particularly cozy in addressing categorical educational data and considerations like missing values or data sparsity. Likelihoods on students' successes and course completions are updated using probabilities based on feature-patterns and class distributions, thus lending the student outcomes a development schedule when one is capable of noticing students likely to get behind for timely interventions and upscaling success.

4. The proposed smart learning optimization model

4.1. Introduction of integrated smart learning architecture

Before we explain our approach, it is appropriate that we present the model which we will be using depicted in Fig. 1. The combined architecture aims to allow transparent monitoring, management and analysis of learning activities in a digital learning environment. At the core is Moodle, a widely used and well-established Learning Management System (LMS) which is well-known for its flexibility and rich functionality in the provision of learning content. To enhance the functionality and interoperability of Moodle [28], we are incorporating two major components: Learning Record Provider (LRP), a CMI-5 compliant Learning Record Store (LRS) and Learning Record Customer (LRC).

Moodle is a foundational platform that facilitates courses, user management, and the sorts to promote engagement between learning material and learners. In order to officially document the numerous learning activities occurring in Moodle, The Learning Record Store (LRS) augments the inherent capability of Moodle by offering a dedicated repository for storing, managing, and analyzing learning experience-related data [29].

The Learning Record Store (LRS) is constructed based on the CMI-5 specifications, ensuring interoperability and compatibility with a vast array of learning platforms and tools. Through single sign-on integration with the Learning Record Provider (LRP), such as Moodle and Assignable Units (AU), the LRS consolidates multiple streams of data, thereby giving educators and administrators a holistic view of learner progress, performance, and behavior.

At the center of Learning Record Store (LRS) functionality is the Assignable Unit (AU) concept, which records single units of learning content or activities. CMI-5-compliant AUs allow for precise measurement of learner activity, skills, and achievements in various learning environments. Teachers can gain a detailed insight into learning capacities by using AUs in the LRS, thereby facilitating data-driven decisions and personalized learning.

4.2. Methodology

The predictive analytics model we use, known as Learning Record Customer (LRC), utilizes supervised machine learning algorithms to predict course duration based on often retrieved data stored in the Learning Record Store (LRS).

The process to be employed for the hybrid system is a structured process in data acquisition, preprocessing, data analysis, and modeling for course completion improvement and learning experience maximization. The steps involved in the implementation of the suggested predictive analytics model in predicting Completion Course Time (CCT) are shown in Fig. 2.

The methodology consists of several key steps:

4.2.1. Identifying data sources

The first step aims to collect the data stored on the LRS and coming

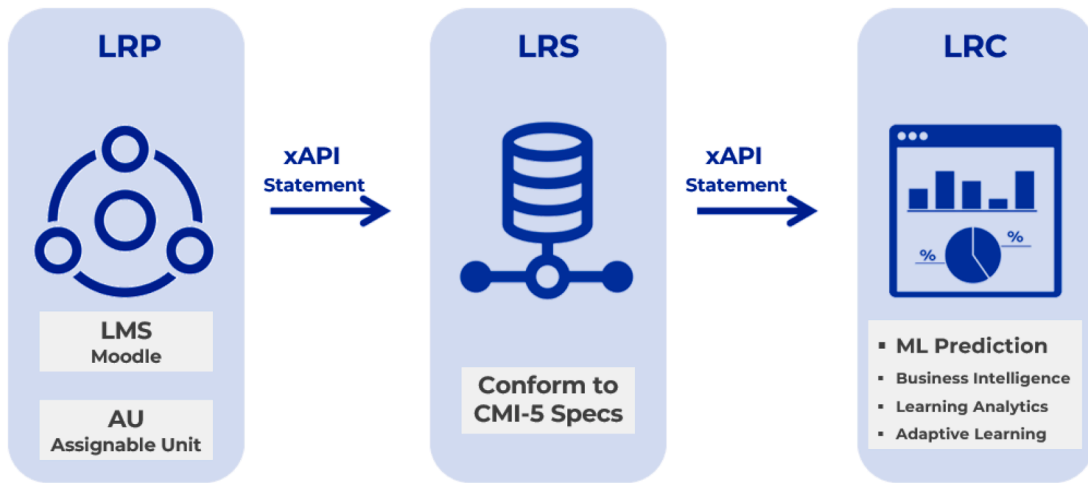


Fig. 1. Suggested model for gathering learning experience data through the inclusion of the CMI-5 specifications.

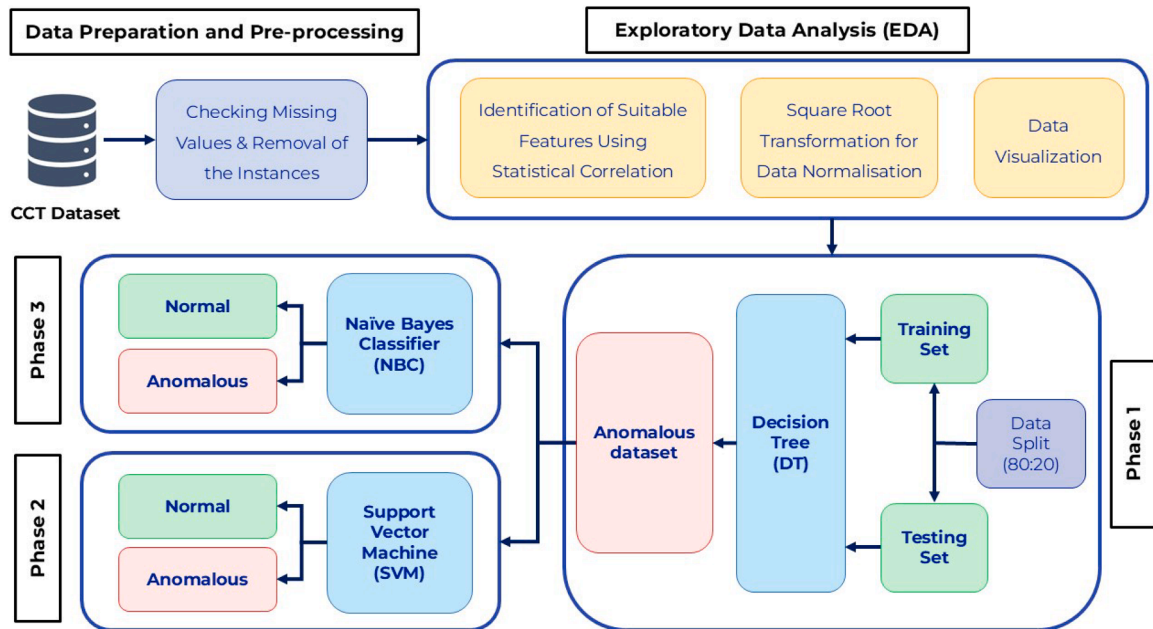


Fig. 2. Proposed framework of hybrid system for improving learning experiences and optimize course completion.

from the LRP via LMS compliant with CMI-5 specifications. These platforms capture a wide range of data points, including learner interactions with course materials, progression through modules, performance on assessments, demographic information, and other relevant variables. The CCT dataset should ideally cover a wide range of courses and learners to ensure the generalizability of the predictive model.

4.2.2. Data preparation and pre-processing

The preprocessing step ensures the quality of the data and gets it ready for analysis after it is collected. Missing value handling, categorical variable encoding, numerical feature normalization, and outlier handling are some examples of these tasks.

We can use the features extracted from the CMI-5 statements to train a machine learning model to predict Completion Course Time. Converting CMI-5 statements shown in Fig. 3 into a machine learning dataset involves parsing JavaScript Object Notation (JSON) structures to extract actor details, actions, resources, results, contexts, and timestamps. These details become features in the dataset (Fig. 3).

As part of the process of creating machine learning models, we first

removed some irrelevant attributes using feature selection within Waikato Environment for Knowledge Analysis (WEKA) latest stable version 3.8 [30]. In order to find the pertinent attributes in the dataset, we choose to apply the WrapperSubsetEval technique for feature selection. This technique employs the Best-first search method. Furthermore, in order to improve predictive performance, feature engineering techniques have been used to generate new variables or modify known ones.

Based on the findings, we eliminated all information pertaining to the other nine attributes in addition to the actor, verb, storage, id, and authority fields; Time spent, progress, completion status, session duration, activity duration, completion threshold, number of interactions, interaction types, and selected metadata in the final dataset. Then, we classified them into five categories as shown in Table 1: Verb-related features, Object features, Contextual features, Interaction features, Additional metadata. and to track the time it takes for students to complete the course, we grouped the different interaction types into: Initialized, Launched, Played, Paused, Interacted, Searched, Terminated, Completed.

```

{
  "actor": {
    "mbox": "mailto:Guest_103.255@elearning.ensad.ma",
    "name": "Guest_103.255",
    "objectType": "Agent"
  },
  "verb": {
    "id": "http://adlnet.gov/expapi/verbs/completed",
    "display": {
      "en-US": "completed"
    }
  },
  "object": {
    "id": "https://www.youtube.com/watch?v=rdguWcRqBBC",
    "definition": {
      "name": {
        "en-US": "YouTube Video"
      },
      "type": "https://w3id.org/xapi/video/activity-type/video"
    },
    "objectType": "Activity"
  },
  "result": {
    "duration": "PT35.17S",
    "completion": true,
    "extensions": {
      "https://w3id.org/xapi/video/extensions/time": 36.219,
      "https://w3id.org/xapi/video/extensions/progress": "0.702",
      "https://w3id.org/xapi/video/extensions/played-segments":
        "1.048[.]5.068[, ]5.068[.]10.108[, ]10.108[.]15.172[, ]15.172[.]36.219"
    }
  },
  "context": {
    "contextActivities": {
      "category": [
        {
          "id": "https://w3id.org/xapi/video"
        }
      ]
    },
    "extensions": {
      "https://w3id.org/xapi/video/extensions/session-id": "571e7b44-2375-4bd4-b37b-0dbbe8b18707",
      "https://w3id.org/xapi/video/extensions/length": 49.683,
      "https://w3id.org/xapi/video/extensions/completion-threshold": "0.70"
    },
    "registration": "eb2d96da-402d-4d79-82ba-1f9a3b77885b"
  },
  "timestamp": "2024-03-08T13:45:05.536Z",
  "id": "64a4d57e-183d-4db3-aeff-eeedb94a0ed2",
  "stored": "2024-03-08T13:45:06.690Z",
  "authority": {
    "account": {
      "homePage": "https://elearning.ensad.ma/xAPI/",
      "name": "42-43a394dec929e1b"
    },
    "objectType": "Agent"
  }
}

```

Fig. 3. Representation JSON of a CMI-5 statement.

4.2.3. Exploratory data analysis (EDA)

To know the nature of the data set and recognize patterns or trends, exploratory data analysis is carried out before predictive models are generated. This comprises plotting the data in the mode of charts, histograms, heatmaps and summary statistics. EDA [34] helps in understanding the distribution of the variables, detecting outliers, and recognizing relationships between features and target variable (CCT).

Tracking student activity for Course Completion Time (CCT) for a data set of 8665 statements between the years 2019 and 2024 displays strong trends in student engagement. With 670 statements and 20.70 min on average time spent and 2.47 interactions, the "Played" action indicates intense engagement with course content. The "Paused" action, with 677 statements, and a mean time spent of 6.87 min with no interactions, indicates short thinking pauses or note-taking. The "Terminated" action, with 218 statements, a mean of 42.22 min, and 3.20 interactions, indicates longer engagement prior to abandonment, possibly with content difficulty. Lastly, the "Completed" action, observed

in 168 statements with a mean length of 36.23 min and a mean of 3.36 interactions, demonstrates the considerable effort and engagement needed to fulfill the course prerequisites and therefore reflects students' persistence in course completion. Tracking of students and an overview of the pattern of their interactions are presented in terms of the overall Course Completion Time (CCT) throughout the dataset presented in Fig. 4.

4.2.4. Proposed framework of hybrid system for improving learning experiences and optimize course completion

The proposed system integrates three algorithms to create a comprehensive model specifically designed to improve learning experiences and optimize course completion time in smart educational environments. Fig. 1 illustrates the process, starting with Phase 1, in which the Decision Tree (DT) algorithm processes the original educational data set to identify important patterns and trends. This first phase helps segment the data based on academic performance and engagement

Table 1
Attributes of the dataset.

Categories	Attributes Name	Descriptions	Types	Values example
Verb-related features	Actor ID	A unique identifier for the actor/user interacting with the course content.	Nominal	userID@domain.edu
	Time spent	The duration of time spent interacting with the course content, typically measured in seconds	Numeric (measured in seconds)	(e.g., 36.464 s)
	Progression	The percentage of completion or progression within the activity. For example, if a video is being watched, progression could indicate how much of the video has been watched.	Numeric (percentage)	70.3 %
	Completion status	Indicates whether the activity has been completed or not. This could be a binary flag (1 for completed, 0 for not completed) or categorical (completed, not completed).	Numeric	Binary flag/ Categorical (1 for Completed or 0 for Not Completed)
	Session duration	The duration of the session during which the interaction took place, if available.	Numeric (measured in seconds)	(e.g., 35.17)
Object features	Activity type	The type of activity being interacted with (e.g., video, course).	Nominal	(Categorical - e.g., Video, Course)
	Activity length	The total length or duration of the activity, typically in seconds.	Numeric (measured in seconds)	(e.g., 49.683 s)
Contextual features	Session ID	A unique identifier for the session during which the interaction occurred.	Nominal (Unique identifier)	"63bb18d8-37af-40f8-911c-eb6a2ab5d6a2"

Table 1 (continued)

Categories	Attributes Name	Descriptions	Types	Values example
Interaction features	Completion threshold	The threshold set for completion of the activity. For example, completion may be considered achieved if progression reaches a certain percentage.	Numeric (Threshold percentage)	(e.g., 70 %)
	Number of interactions	The total number of interactions made during the session.	Numeric (Count)	(e.g., 5)
	Types of interactions	Specific types of interactions made during the session, if available.	Nominal	(Categorical - e.g., Initialized, Launched, Played, Paused, Searched, Terminated, Completed)
Additional metadata	Metadata	Any additional information or metadata provided in the CMI5 statements that might be relevant for predicting completion course time. This could include information about the user, context, or environment.	Nominal	(Various metadata fields such as Age: 25-30, Gender: Female, Browser: Chrome v 94.0, Location: Casablanca, Morocco)

metrics. The next phase uses Support Vector Machines (SVM) to analyze the filtered data from Phase 1. SVMs excel as binary classifiers in this context, effectively distinguishing between students who are on track to successfully complete their course and those who may need additional support. Using kernel functions, SVMs process complex, nonlinear relationships in educational data, improving prediction accuracy. The goal of SVM is to find a hyperplane that maximizes the space between classes [31], which is formulated as follows:

$$\min_{w,b,\xi} \left\{ \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \xi_i \right\} \quad (6)$$

subject to:

$$y_i(w \cdot \phi(x_i) + b) \geq 1 - \xi_i \quad (7)$$

$$\xi_i \geq 0$$

where ξ_i are the slack variables, b is the bias, w is the weight vector, and $\phi(x)$ is the kernel function mapping x to a higher-dimensional space.

While such a model distinguishes well between students who complete a course without problems and those at risk of failing it, they recursively split the dataset into subsets based on feature values, usually capturing complicated relationships in the educational database. Structured like a tree, the Decisioning Tree enhances prediction accuracy substantially. Each decision within a Decision Tree makes a

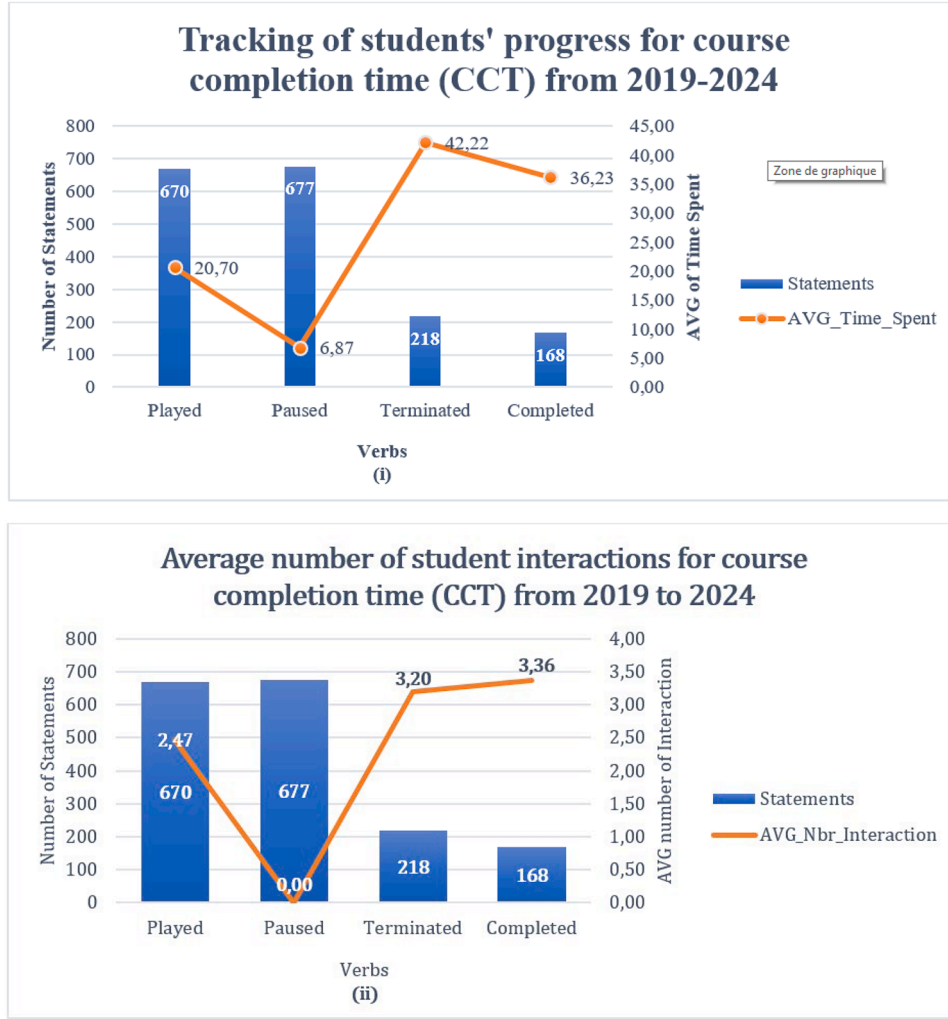


Fig. 4. Visualizing (i) the Tracking of students' progress and (ii) their interactions for course completion time (CCT) from 2019 to 2024 for Dataset = 8665 statements.

partition of feature space such that class separation always takes place through a node with discrete probabilities.

Apart from being a suitable idea of classification, the general form of a Decision Tree model can be represented as a sequence of binary decisions based on feature values that generate the class outcome. The optimization goal is to minimize classification error [32], which can be presented as:

$$\min_T \sum_{i=1}^n \text{Loss}(y_i, \hat{y}_i) \quad (8)$$

where T represents the tree structure, that is, the splits and the decisions made at each node. $\text{Loss}(y_i, \hat{y}_i)$ is the loss function that measures the difference that is between the true class label y_i and the predicted label of $\hat{y}_i$.

Phase 3 involves training a Naive Bayesian Classifier (NBC) with the filtered dataset from the previous stages to provide probabilistic insights into student outcomes. NBC simplifies probability calculations by assuming conditional independence among features [33], represented by:

$$P(C_k|x) = \frac{P(C_k) \prod_{i=1}^n P(x_i|C_k)}{P(x)} \quad (9)$$

where $P(C_k)$ is the prior probability of class C_k , $P(x_i|C_k)$ is the likelihood of feature x_i given class C_k , and $P(x)$ is the marginal likelihood of the

data.

This ensemble system of DT-SVM-NBC utilizes the strengths of each model to optimize prediction accuracy and efficiency. Decision Trees construct an understandable framework through data partitioning based on performance thresholds, making initial feature extraction and data division simpler. SVMs refine these divisions by identifying complex patterns and outliers in high-dimensional space. NBC then provides probability scores to help differentiate between successful and failing students. Hybridizing these algorithms improves detection quality and sensitivity to changing learning contexts and data characteristics. The method balances between accuracy and efficiency by appropriate parameter tuning and computational architecture. DT-SVM-NBC hybrid system integrates the individual strengths of decision trees, support vector machines (SVMs), and Naive Bayes classifiers to deliver a powerful and well-rounded system to enhance learning experience and ensure course completion in smart education contexts [35,36].

5. Experimental analysis and evaluation

For the study, we used an Intel Core i5-9300 with 2.40 GHz and 16 GB of RAM. Implementation of models was done using Python with the addition of Python machine learning libraries including Sklearn and Numpy libraries. The model is divided into three stages with training of Decision Trees (DT), Support Vector Machines (SVM) and the Naive Bayes Classifier (NBC) being included in the first, second and third

stages, respectively. For DT algorithm: We have used the information gain approach and applied the percentage approach of performance measure (80 % training data set and 20 % test data set). In the second stage, we used SVM with polynomial kernels. The performance of the Naïve Bayes algorithm was evaluated using 10-fold cross-validation in the third phase. To measure the classification performance, several loss, score and utility functions from the sklearn.metrics module were used. Metrics often require confidence values, binary decision values, or positive class probability estimates. In most implementations, each sample can contribute weighted to the final result via the sampling weight argument.

The measures more commonly used in the studies evaluated are listed below. These actions are based on the expected/Boolean anomaly labels applied to a specific data item. Algorithms typically provide a numerical anomaly score that indicates the degree of anomaly of a particular data point compared to others, rather than a simple true or false answer.

The following measures are used in this study:

- **Precision (P):** The ratio of True Positives (TP) to the sum of False Positives (FP) and True Positives (TP). True Positives are detected anomalies that correlate with attack manifestations, while False Positives are detected anomalies that do not.

$$Precision = \frac{TP}{(TP + FP)} \quad (10)$$

- **Recall (R):** Often stated alongside Precision, Recall quantifies the ratio of True Positives (TP) to the sum of TP and False Negatives (FN), which are undetected anomalies.

$$Recall = \frac{TP}{(TP + FN)} \quad (11)$$

- **F-Measure and F-Score (F):** Metrics to evaluate the performance of the classification model. The F-Score combines Precision (P) and Recall (R) using one parameter. If this parameter is greater than 1, Recall is weighted more heavily than Precision. The F-Measure, particularly F1, is used when False Positives (FPs) and False Negatives (FNs) are considered equally undesirable. The balanced mean of P and R is the arithmetic average.

$$F1 - score = 2 * \frac{(Precision * Recall)}{(Precision + Recall)} \quad (12)$$

- **Accuracy:** Refers to the proportion of correct detections, including both True Positives and True Negatives, out of the total number of data points evaluated. This combines positive and negative reviews into a single statistic.

$$Accuracy = \frac{(TP + TN)}{(TP + TN + FP + FN)} \quad (13)$$

- **Area Under the Receiver Operating Characteristic Curve, or ROC Curve (AUC):** Shows the performance of binary classifiers when their discrimination thresholds vary, visualized by plotting recall versus false positive rate. A high AUC value indicates a technique that is often acceptable for the target data set.

$$ROC - AUC = \frac{TPR}{FPR} \quad (14)$$

Among the models examined, DT-SVM-NBC performs exceptionally

well in terms of accuracy, precision, recall and F1-score. With an accuracy of 99.94 %, DT-SVM-NBC demonstrates superior accuracy in correctly classifying data as normal or abnormal. This accuracy is significantly higher than that of the Decision Trees, Support Vector Machines and Naïve Bayes Classifier models, which achieve accuracies of 85.60 %, 78.24 % and 82.40 %, respectively (Table 2).

A comparative analysis with existing literature highlights the superior performances of our proposed DT-SVM-NBC model in a multiplicity of metrics of assessment: precision, recall, and F1-score. With respect to this function, the DT-SVM-NBC model has an outstanding precision score of 99.99 %, decidedly outperforming Decision Trees (84.90 %), Support Vector Machines (79.65 %), and Naïve Bayes Classifier (81.25 %), indicating a higher efficiency for the model in the detection of anomalous data while having less tolerance to false positives, which is pivotal for a smart learning analytics reliability.

The DT-SVM-NBC, with recall at 99.94 %, can identify just about all true anomalous cases within the dataset. This far exceeds the performance in recall of Decision Trees (88.75 %), Support Vector Machines (75.50 %), and Naïve Bayes Classifier (84.55 %), which suggests DT-SVM-NBC is superior in finding anomalies with precious few omissions (Fig. 5).

Given the F1-score captures a balance between precision and recall, this further propounds the DT-SVM-NBC model with remarkable 99.96 % being higher than that of Decision Trees (86.79 %), Support Vector Machines (77.55 %), and Naïve Bayes Classifier (82.87 %), this demonstrates the robustness of the model in classification of anomalous data even while suppressing the number of false positives it produces (Fig. 5).

The DT-SVM-NBC hybrid model shows paramount importance, for it beats all individual classifiers across all common evaluation metrics. By combining the strengths of both Decision Trees and SVMs, the models make the best out of both methodologies, attaining greatly enhanced accuracy concerning classification, anomaly detection, and optimization potential. This hybrid provides a more enabling and trustworthy framework for spotting outliers in smart learning environments, which in turn provides DT-SVM-NBC a leg up on building sustainable and robust performance in high-precision and recall per model.

6. Conclusion

The exponential growth of data in various fields, including education, finance, and healthcare, has spurred the development of advanced algorithms to optimize large data environments. Identifying significant deviations from expected patterns or optimizing performance is critical to uncovering trends that could indicate inefficiencies, anomalies, or other critical issues. The goal of this effort was to explore the application of optimization techniques in large-scale data environments. We have developed a Smart Learning Optimization Model that leverages machine learning to improve performance and accuracy. The proposed system uses a hybrid approach that integrates Decision Trees (DT), Support Vector Machines (SVM) and Naïve Bayes Classifiers (NBC). By combining the unique strengths of each model, this hybrid system improves processing efficiency and recognition accuracy in educational environments. Decision trees are valued for their ability to structure and partition data efficiently, SVMs excel at managing high-dimensional spaces and complex patterns, and NBC offer simplicity and effective

Table 2
Performance comparison of smart learning optimization models.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
DT-SVM-NBC	99.94	99.99	99.94	99.96
Decision Trees	85.60	84.90	88.75	86.79
Support Vector Machines	78.24	79.65	75.50	77.55
Naïve Bayes Classifier	82.40	81.25	84.55	82.87

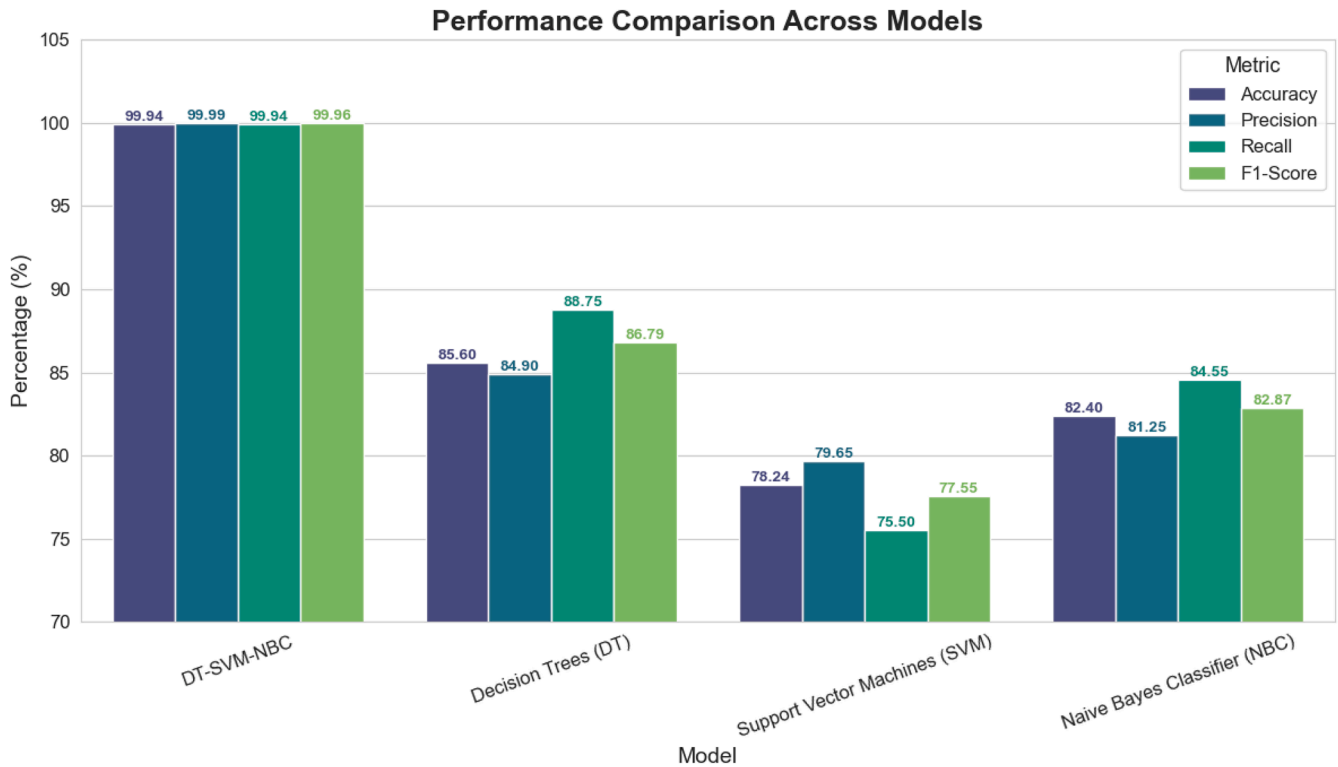


Fig. 5. Comparison of Model Performance Metrics.

probabilistic reasoning.

Our approach shows that this integration significantly increases the optimization possibilities. Furthermore, we used a real dataset to conduct a comprehensive comparison between our proposed technique and existing methods. The results show that the DT-SVM-NBC hybrid model outperforms traditional models and has superior accuracy, precision, recall and F1 score. This suggests that our model provides a robust solution for optimizing performance in big data environments and provides valuable insights for the field of intelligent education and beyond.

List of acronyms

NBC, NAIVE BAYESIAN CLASSIFIER
 SVM, Support Vector Machine
 DT, Decision Tree
 KL, Kullback-Leibler
 AI, Artificial Intelligence
 ML, Machine Learning
 LMS, Learning Management System
 LRP, Learning Record Provider
 LRS, Learning Record Store
 LRC, Learning Record Customer
 AU, Assignable Unit
 CMI-5, Communication and Media Interoperability-5
 CCT, Completion Course Time
 WEKA, Waikato Environment for Knowledge Analysis
 EDA, Exploratory Data Analysis

Funding

The authors received no financial support for the research, authorship, and/or publication of this article.

Statements on open data and ethics

Data can be provided upon a justified request.

Declaration of generative AI and AI-assisted technologies in the writing process

The authors in the current study did not use any kinds of artificial intelligence in any parts, and sections of the article.

Ethics approval and consent to participate

Not applicable. This study did not involve human or animal participants. Only behavioral data from a distance learning system was used.

Consent for publication

Not applicable. This manuscript does not contain any individual person's data in any form.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

CRediT authorship contribution statement

Abdellah Bakhoui: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Formal analysis, Conceptualization. **Amine Dehbi:** Conceptualization. **Lahcen Amhaimar:** Writing – review & editing. **Said Broumi:** Writing – review & editing. **Mohamed Talea:** Writing – review & editing, Investigation, Formal analysis. **Abderrahim Khalidi:** Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors are highly grateful to the Editor-in-Chief, Professor Ke Xue, and the referees for their valuable comments and suggestions.

We thank the direction of Information System of Hassan II University of Casablanca and Information System Committee (ISC) at National Higher School of Art and Design (UH2C) for providing the "i-UH2C" datasets and they give us the permission to use them to carry out this work.

Data availability

Data will be made available on request.

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